

Literature Status: The Real-Line Gaussian Tail-Space Subcase of Question 7

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Status. Literature implied answer, real-line subcase. This is not a new proof packet.

Source Question

The source paper is A. Eskenazis and P. Ivanisvili, *Dimension independent Bernstein–Markov inequalities in Gauss space*, arXiv:1808.01273v3, Journal of Approximation Theory 253 (2020), Article 105377.

Question 7 asks whether, for every $1 < p < \infty$, there is $c_p > 0$ such that for every $k \geq 1$, every $d \geq 1$, and every polynomial

$$P(x) = \sum_{|\alpha| \geq d} c_\alpha H_\alpha(x)$$

on $\mathbb{R}^k$, one has

$$\|LP\|_{L^p(d\gamma_k)} \geq c_p d \|P\|_{L^p(d\gamma_k)}.$$

Here $L = \Delta - x \cdot \nabla$ is the Ornstein–Uhlenbeck generator, with $LH_\alpha = -|\alpha|H_\alpha$ in the source convention.

Supporting Paper

The supporting paper is A. Eskenazis and P. Ivanisvili, *Sharp growth of the Ornstein–Uhlenbeck operator on Gaussian tail spaces*, arXiv:2011.01359v2; Israel Journal of Mathematics 253 (2023), 469–485, DOI 10.1007/s11856-022-2373-8.

In the introduction, Corollary 2 states that for every $p \in (1, \infty)$ there are constants $S_p, T_p < \infty$ such that, for every one-dimensional polynomial f in the d -tail space,

$$d\|f\|_{L^p(d\gamma_1)} \leq T_p \|Lf\|_{L^p(d\gamma_1)}.$$

Equivalently,

$$\|Lf\|_{L^p(d\gamma_1)} \geq T_p^{-1} d \|f\|_{L^p(d\gamma_1)}.$$

This is precisely the $k = 1$ case of Question 7.

Scope Limitation

The supporting result answers only the real-line subcase of the source question. It does not settle Question 7 for arbitrary k -dimensional Hermite-tail polynomials, nor Question 3 on the sharp $\sqrt{\deg P}$ gradient Bernstein–Markov inequality in all dimensions.

Provenance

The supporting authors explicitly know the relation: arXiv:2011.01359 says the second inequality answers the Gaussian analogue of Mendel–Naor’s question on the real line and cites the source paper arXiv:1808.01273. Because the original Question 7 is stated for all $k \geq 1$, this packet is classified as `literature_implied_answer (real-line subcase)`, not as a full literature answer.